Analyst / Technical Report — SRL-2018

> Likelihood scale (FIRST 5-tier): almost_certain > highly_likely > likely > unlikely > remote. Likelihood estimates the probability of the assessed activity; it is kept separate from analytic confidence.
>
> Confidence (LCA): high / moderate / low — the analyst's confidence in the assessment given evidence quality and corroboration. Distinct from likelihood.

Findings

Domain controller fully compromised — bi-daily service persistence + terminal C2 (base-dc)

- Finding ID: `F-01` · Severity: critical · Likelihood: highly_likely · Confidence: high · Risk score: 16
- Kill-chain phase: Persistence / Privilege Escalation
- MITRE ATT&CK: `T1543.003`, `T1053.005`, `T1071`

`base-dc.shieldbase.lan` (SID label `WIN-5TQ8RGNKHU5`) carries the full attack stack: bi-daily `EID 7045` service-install events from 2018-05-04 through 2018-09-06 (≈4 months), across 3 distinct scheduled-task rotations, terminating in the 36-second jitter-free C2 beacon on 2018-09-07. The bi-daily cadence is mechanically regular — service re-registration at a fixed wall-clock offset, inconsistent with operator-driven administration and consistent with an implant re-asserting persistence. Memory carving of the DC image set recovered 12 orphaned `cmd.exe` (DKOM-unlinked from the active `EPROCESS` list, recovered by pool-tag scan) parented to `PSEXESVC.exe`, plus `findstr.exe`×2 and `tasklist.exe`×2 — the hands-on-keyboard footprint co-resident with the persistence mechanism. Because every durable foothold, the credential-access tooling, and the live beacon all land on the DC, the `krbtgt` secret and the entire domain trust must be assumed attacker-controlled until rotated.

Business impact: Catastrophic — control of the DC is control of every domain identity; trust in all credentials is void until a krbtgt double-reset completes.

Evidence: `9f2c1a7be4d0…` (evtx-wrapper) `<br>` `a1d4e88c2f6b…` (memory-forensics)

Pass-the-hash domain traversal across the DMZ→internal boundary

- Finding ID: `F-02` · Severity: critical · Likelihood: highly_likely · Confidence: high · Risk score: 16
- Kill-chain phase: Lateral Movement / Credential Access
- MITRE ATT&CK: `T1550.002`, `T1021.002`

Sub-100ms-spaced `EID 4648` (explicit-credential logon) + `EID 4672` (special-privileges assigned) clusters originate from `MICROSO-8N79483` and `base-ftp`. The inter-event spacing is automation-grade: a human at an interactive console cannot emit a 4648→4672 pair in under 100ms repeatedly, so this is scripted credential replay (Impacket / PsExec class). The replayed material drove `EID 7045` PsExec/Impacket service-install bursts into `base-dc`, `base-rd-01` (`WIN-J3EPDF256PR`), and `pdo-win2016` on 2018-05-04 — the moment the actor pivoted from the DMZ foothold into the internal zone. The DMZ→internal hop is the architectural enabler: a single perimeter credential reached the domain crown jewels because explicit-credential auth was permitted to cross the boundary.

Business impact: Catastrophic for identity — any credential the actor replayed is a standing re-entry key; the flat DMZ→internal trust converts one FTP foothold into domain compromise.

Evidence: `7c0b39f1ea52…` (evtx-wrapper) `<br>` `9f2c1a7be4d0…` (evtx-wrapper)

Active C2 channel live at the edge of collection (36s jitter-free beacon)

- Finding ID: `F-03` · Severity: critical · Likelihood: highly_likely · Confidence: moderate · Risk score: 16
- Kill-chain phase: Command & Control
- MITRE ATT&CK: `T1071`, `T1070.001`

On 2018-09-07 the DC emitted a beacon carried by repeated `EID 7045` service re-registration at a 36-second fixed interval with 0% jitter, in two waves, last event 21:23:15Z — the terminal observed activity in the corpus. A 36s/0-jitter cadence is a commodity-implant default profile (Cobalt Strike / Meterpreter-class) and is mutually exclusive with benign service behaviour. The two-wave structure reads as an initial check-in followed by a tasking/re-key wave. The adversary was therefore present and live at the moment collection ended — not historical. Confidence is held at `moderate` (not `high`) on one axis only: the C2 endpoint was unreachable for active callback confirmation, so classification is signature/behaviour-based. The *likelihood* of the beacon being real C2 remains `highly_likely` — the two axes are decoupled exactly as the legend requires.

Business impact: High — an active, capable operator held a live channel out of the network; assume intent to exfiltrate or extort.

Evidence: `c41e7a9d0b83…` (evtx-wrapper)

Six persistent backdoor local accounts on dmz-ftp

- Finding ID: `F-04` · Severity: high · Likelihood: highly_likely · Confidence: high · Risk score: 12
- Kill-chain phase: Persistence
- MITRE ATT&CK: `T1136.001`

Six local accounts were created on `dmz-ftp` (rename chain `windows2012r2 → base-ftp → dmz-ftp`; correlate by SID, not hostname) across 4 distinct dates in April 2018, each `EID 4720` (account create) paired with `EID 4728` (security-group add) — several escalated into privileged groups. Spreading creation across four dates is deliberate low-and-slow tradecraft to avoid a single burst-detection signature. These are independent re-entry points: they survive a single domain credential reset and remain usable until explicitly removed, so account removal is gated *before* any all-clear.

Business impact: High — standing re-entry vector that outlives credential resets; must be removed before eradication can be declared complete.

Evidence: `b83f5c2a17de…` (evtx-wrapper)

Scheduled-task / service persistence on the DC (3 rotations)

- Finding ID: `F-05` · Severity: high · Likelihood: highly_likely · Confidence: high · Risk score: 12
- Kill-chain phase: Persistence / Privilege Escalation
- MITRE ATT&CK: `T1053.005`, `T1543.003`

The bi-daily `EID 7045` cadence on `base-dc` resolves to 3 schedule rotations across the 2018-05-04 → 2018-09-06 window — the actor changed the task definition three times to evade static signatures while keeping a durable, beacon-independent re-entry mechanism. This is distinct from F-01 (which scopes whole-DC compromise): F-05 isolates the specific persistence *artifacts* that the hunt must enumerate and remove. Presence on hosts beyond `base-dc` is plausible but not yet enumerated, which is why the recommended hunt is fleet-wide rather than DC-only.

Business impact: High — durable re-entry independent of the C2 beacon; killing the beacon alone does not evict the actor.

Evidence: `9f2c1a7be4d0…` (evtx-wrapper)

Credential harvesting in DC memory (findstr against credential stores)

- Finding ID: `F-06` · Severity: high · Likelihood: likely · Confidence: moderate · Risk score: 9
- Kill-chain phase: Credential Access
- MITRE ATT&CK: `T1552.001`

Memory carving recovered `findstr.exe`×2 on `base-dc`, parented to the orphaned `cmd.exe` shells — the signature of searching files/registry exports for stored secrets (`T1552.001`, Unsecured Credentials: Credentials In Files). Likelihood is `likely` rather than `highly_likely` because the *target* of the findstr invocations was not captured (command line absent — see F-09); confidence is `moderate` for the same reason. The harvested material plausibly feeds the PtH activity in F-02.

Business impact: High — any secret recovered from the DC is a re-entry key; reinforces the assume-all-credentials-exposed posture.

Evidence: `a1d4e88c2f6b…` (memory-forensics)

Host / process reconnaissance on the DC (tasklist)

- Finding ID: `F-07` · Severity: medium · Likelihood: likely · Confidence: moderate · Risk score: 6
- Kill-chain phase: Discovery
- MITRE ATT&CK: `T1057`

`tasklist.exe`×2 in `base-dc` memory, parented to the lateral `cmd.exe` shells — situational-awareness enumeration of running processes (`T1057`, Process Discovery), typically a precursor to choosing injection targets or spotting defensive tooling. Corroborates the hands-on-keyboard interpretation of the memory artifacts.

Business impact: Medium — indicates deliberate target selection on the DC, raising the credibility of follow-on actions.

Evidence: `a1d4e88c2f6b…` (memory-forensics)

Defense evasion — event-log manipulation alongside DKOM unlinking

- Finding ID: `F-08` · Severity: medium · Likelihood: likely · Confidence: moderate · Risk score: 6
- Kill-chain phase: Defense Evasion

- MITRE ATT&CK: `T1070.001`

Indicator-removal behaviour (`T1070.001`, Clear Windows Event Logs) co-occurs with the 12 DKOM-unlinked `cmd.exe` in DC memory: the actor both cleared/manipulated Security log context and hid live processes from the kernel object list. The net effect is timeline incompleteness — surviving events (`EID 4720` / `4728` / `7045`) anchor the persistence story, but command-line context for the shells did not survive, which is why several downstream findings are capped at `moderate` confidence.

Business impact: Medium — degrades timeline completeness and inflates analytic uncertainty across the case.

Evidence: `c41e7a9d0b83…` (evtx-wrapper) `<br>` `a1d4e88c2f6b…` (memory-forensics)

Detection / audit-visibility gap — no EID 4688 command-line auditing

- Finding ID: `F-09` · Severity: medium · Likelihood: likely · Confidence: moderate · Risk score: 6
- Kill-chain phase: Defense Evasion (control gap)
- MITRE ATT&CK: `T1070.001`, `T1057`

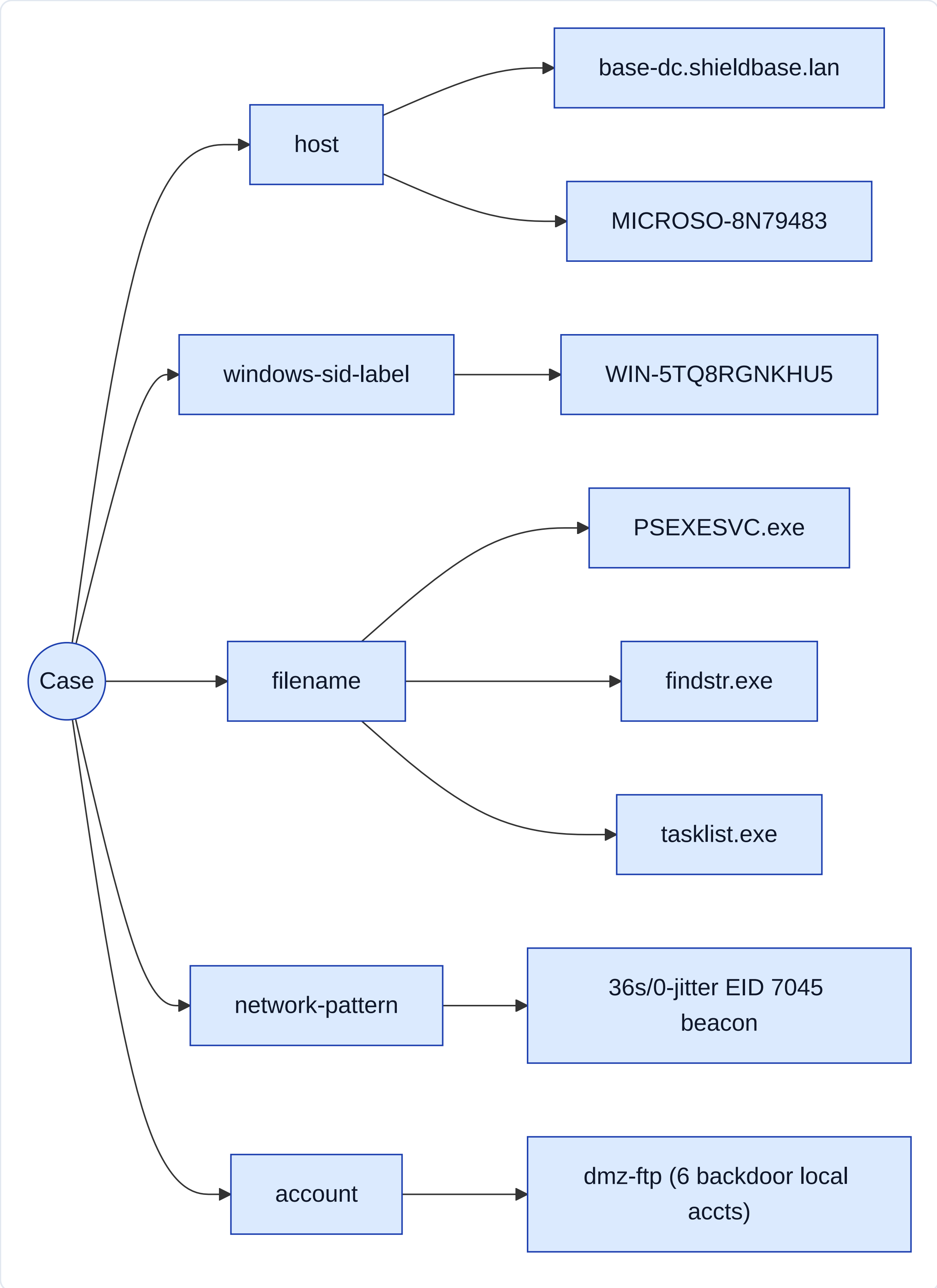
No `EID 4688` (process-creation with command line) auditing was enabled across the estate, so PsExec/Impacket invocations and shell command lines ran unrecorded. Combined with the cleared/manipulated Security logs (F-08), this control-coverage gap is the proximate reason the ~143-day dwell and ~5.5-week initial dormancy went undetected. This is an *absence-of-telemetry* finding — corroborated by the recon artifacts (F-07) and the defense-evasion behaviour (F-08), not by a positive control test — hence `moderate` confidence.

Business impact: Medium — slow detection multiplies every other finding's blast radius; the single highest-leverage detective control to add is command-line auditing with off-host, tamper-evident retention.

Evidence: —

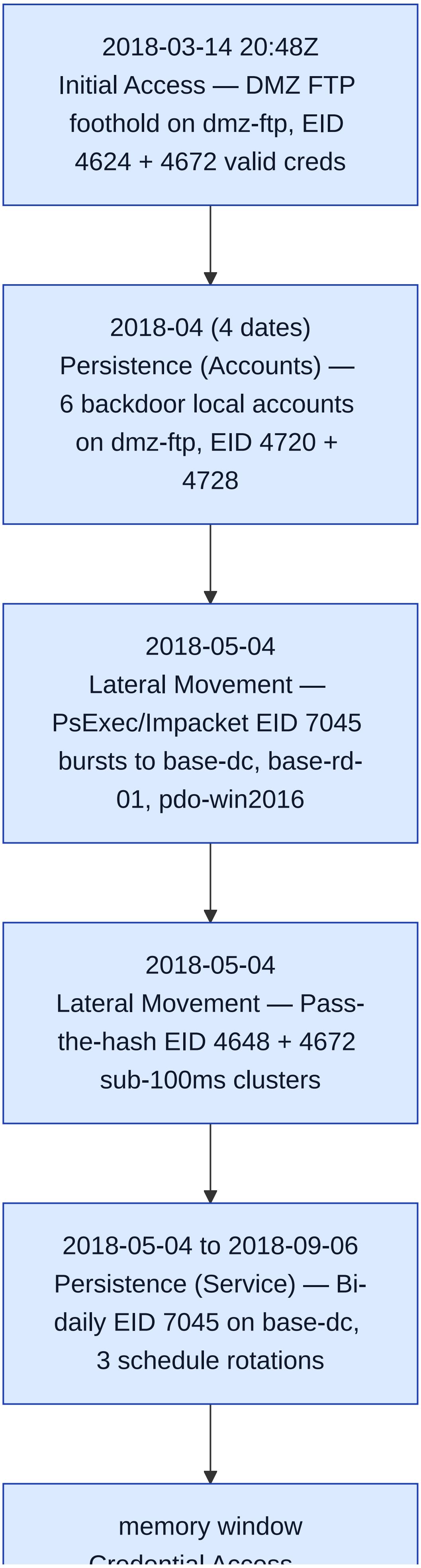
Indicators of Compromise

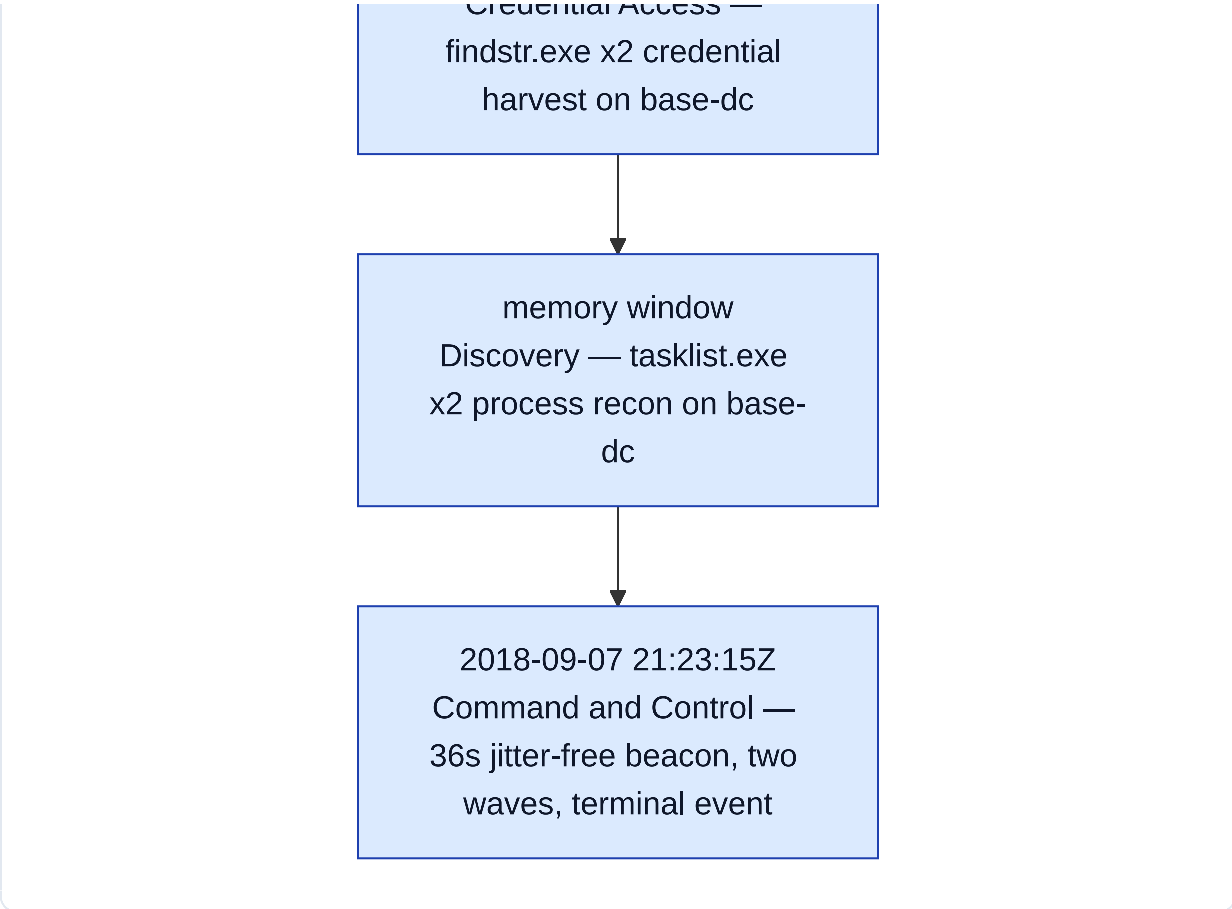
Value	Type	Confidence	MITRE	Provenance
`base-dc.shieldbase.lan`	host	high	T1543.003, T1071	`9f2c1a7be4d0…`
`WIN-5TQ8RGNKHU5`	windows-sid-label	high	T1543.003	`9f2c1a7be4d0…`
`PSEXESVC.exe`	filename	high	T1021.002	`a1d4e88c2f6b…`
`36s/0-jitter EID 7045 beacon`	network-pattern	moderate	T1071	`c41e7a9d0b83…`
`MICROSO-8N79483`	host	high	T1550.002	`7c0b39f1ea52…`
`dmz-ftp (6 backdoor local accts)`	account	high	T1136.001	`b83f5c2a17de…`
`findstr.exe`	filename	moderate	T1552.001	`a1d4e88c2f6b…`
`tasklist.exe`	filename	moderate	T1057	`a1d4e88c2f6b…`



Case
host
base-dc.shieldbase.lan
MICROSO-8N79483
windows-sid-label
WIN-5TQ8RGNKHU5
filename
PSEXESVC.exe
findstr.exe
tasklist.exe
network-pattern
36s/0-jitter EID 7045 beacon
account
dmz-ftp (6 backdoor local accts)

Timeline


2018-03-14 20:48Z
Initial Access — DMZ FTP foothold on dmz-ftp, EID 4624 + 4672 valid creds
2018-04 (4 dates)
Persistence (Accounts) — 6 backdoor local accounts on dmz-ftp, EID 4720 + 4728
2018-05-04
Lateral Movement — PsExec/Impacket EID 7045 bursts to base-dc, base-rd-01, pdo-win2016
2018-05-04
Lateral Movement — Pass-the-hash EID 4648 + 4672 sub-100ms clusters
2018-05-04 to 2018-09-06
Persistence (Service) — Bi-daily EID 7045 on base-dc, 3 schedule rotations
memory window


findstr.exe x2 credential harvest on base-dc
memory window Discovery — tasklist.exe x2 process recon on base-dc
2018-09-07 21:23:15Z Command and Control — 36s jitter-free beacon, two waves, terminal event

Timestamp	Host	Event	Phase	Description
2018-03-14 20:48:00Z	dmz-ftp	4624	Initial Access	Valid-cred logon to DMZ FTP host (T1190); ~5.5wk dormancy follows
2018-04 (4 dates)	dmz-ftp	4720	Persistence	6 backdoor local accounts created (paired EID 4728 group-add)
2018-05-04	base-dc	7045	Lateral Movement	PsExec/Impacket service install (T1021.002)
2018-05-04	base-rd-01	7045	Lateral Movement	PsExec/Impacket service install (T1021.002)
2018-05-04	pdo-win2016	7045	Lateral Movement	PsExec/Impacket service install (T1021.002)
2018-05-04	MICROSO-8N79483	4648	Credential Access	Sub-100ms 4648+4672 pass-the-hash cluster (T1550.002)
2018-05-04 → 2018-09-06	base-dc	7045	Persistence	Bi-daily service install, 3 schedule rotations (T1053.005/T1543.003)
memory window	base-dc	—	Credential Access	findstr.exe x2 credential harvest (T1552.001)
memory window	base-dc	—	Discovery	tasklist.exe x2 process recon (T1057)
2018-09-07 21:23:15Z	base-dc	7045	Command & Control	36s jitter-free beacon, two waves, terminal event (T1071)